MIL-C-51448(EA)
30 June 1978
(See section 6)

MILITARY SPECIFICATION

CANISTER

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of canister assembly for 2.75 inch smoke warhead.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition, General Specification For.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1168 - Lot Numbering of Ammunition.

FSC 1340

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

D90-1-301 - Canister.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

UNIFORM FREIGHT CLASSIFICATION

(Application for copies of these ratings, rules, and regulations should be addressed to Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606).

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing D90-1-301 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the canister assembly shall conform to the specifications and drawings listed on Drawing D90-1-301 and subsidiary drawings.

3.2 Assembly. The canister shall be assembled in accordance with Drawing D90-1-301. The canister assembly shall pass through the ring gage specified on Drawing D90-1-301.

3.3 Leakage. The canister assembly shall not leak when subjected to an internal air pressure of 75 to 100 pounds per square inch (psi) (517 to 689 kilopascals) at ambient temperature when tested as specified in 4.4.4.1.

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3.4 Strength. The canister assembly shall not burst when subjected to a minimum hydrostatic pressure internally of 300 psi (2068 kilopascals) when tested as specified in 4.4.4.2.

3.5 Preproduction. Prior to the start of regular production, a preproduction sample of the canister assemblies shall be produced in accordance with this specification for examination and tests (see 4.3).

3.6 Workmanship.

3.6.1 Materials. All materials shall be free from dirt, rust, corrosion, grease, chips or other foreign matter. Materials shall be free from extraneous porosity, cavitation, warp, laps, cracks, inclusions, pits, voids or other injurious defects.

3.6.2 Parts and assemblies. All parts and assemblies shall be sound, of uniform quality and condition and free from seams, cracks, nicks, warp, obstructions, inclusions, scratches, dents, scrapes, gouges, holes, voids, porous areas, distortion, laps, pits, rust, corrosion, or other injurious defects or blemishes. Cleaning shall be thorough to remove all dirt, chips, grease, oil, flux, residues, chemical deposits or other foreign matter; and shall not be injurious to any part nor shall parts be contaminated by the cleaning agents.

4. QUALITY CONFORMANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as specified herein, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Additional contractor's responsibility. Drawings for contractor designed inspection equipment (other than standard measuring equipment) shall be forwarded to the design agency for approval prior to use for production (see 6.2).

4.1.3 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of section 3 and section 5, for which specific inspection has not been provided in this specification have been satisfied.

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4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Lot. A preproduction lot of 50 each canister assemblies, Drawing D90-1-301 and 50 each bodies Drawing D90-1-298 shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production. Samples as required by 4.3.2 shall be forwarded to the design agency or Government designated laboratory for inspection and evaluation prior to proceeding with regular production (see 6.2). Provision of MIL-A-2550 shall apply.

4.3.2 Inspection procedure.

4.3.2.1 For examination and nondestructive tests. The sample canister assemblies and canister bodies shall be examined and tested in accordance with the classification of defects (4.4.3.4). Sampling shall be in accordance with MIL-STD-105, inspection level II.

4.3.2.2 For destructive test. Five of the sample canisters shall be tested in accordance with 4.4.4.2. No failures are allowed.

4.3.3 Acceptance/rejection criteria. The preproduction sample shall comply with the examinations and tests specified in 4.3.2 to be acceptable. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. Each lot shall be controlled in accordance with MIL-STD-1168. A lot shall consist of like components or like subassemblies of one interfix number from one manufacturer, from the same materials and under essentially the same manufacturing conditions.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive test. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive tests. Two canister assemblies shall be taken at random from each eight hour shift (see 4.4.3.2).

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4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive test. The sample canister assemblies shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

4.4.3.2 For destructive tests. The sample canister assemblies shall be tested in accordance with 4.4.4.2. No failures are allowed.

4.4.3.3 For critical defects. Each item in the lot shall be tested and examined for the critical characteristics listed in the classification of defects.

4.4.3.4 Classification of defects. See pages 6 and 7.

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Leakage. With the canister assembly submerged in water, and using a clean, dry air pressure source, apply the specified pressure through the plug end of the canister for a minimum of 15 seconds, watching for any bubbles which would indicate a leak.

4.4.4.2 Strength. Subject the canister assembly to the specified hydrostatic pressure using a suitable fixture for a minimum of 10 seconds, then increase pressure until rupture occurs and record pressure.

5. PACKAGING

5.1 Packaging, interplant shipment. The canister assembly shall be packaged to provide adequate protection against deterioration and damage resulting from transit from the supply source to the first receiving activity.

5.2 Packing, interplant shipment. The canister assembly, packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with Uniform Freight Classification rules or the regulations of other common carriers applicable to the mode of transportation.

5.3 Marking. In conjunction with information furnished in the contract, shipping containers shall be marked in accordance with MIL-STD-129 as applicable.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.3.4a	Canister				D90-1-301
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY D90-1-302 PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical:					
1	Leakage		100%	3	4.4.4.1
Major:					
101	Outside diameter incorrect		0.40%	3	D90-1-301 Note 8 (Gage)
102	Overall length incorrect		0.65%	3	Gage/SMI
103	Burster threads incorrect		0.65%	3	Gage
104	Inside diameter of burster tube incorrect		0.65%	3	Gage
Minor:					
201	Paint damaged (as applicable)				
202	Workmanship		0.65%	3	Visual
	Note: SMI - Standard measuring instrument.				

NOTES:

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.3.4b	Canister				C90-1-298 Canister Body
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY D90-1-301 PARAGRAPH REFERENCE/ INSPECTION METHOD
Critical:	None defined				
Major:					
101	Outside diameter incorrect		0.65%	3	Gage
102	Inside diameter incorrect		0.65%	3	Gage/SMI
103	0.750 dimension incorrect		0.65%	3	SMI
104	Groove incorrect		0.40%	3	SMI
105	Wall thickness incorrect		0.65%	3	SMI/Gage
106	Dimension from end of canister to point where groove depth is less than min depth specified (0.020)		0.65%	3	SMI
	NOTE: SMI - Standard measuring instrument.				

NOTES:

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6. NOTES

6.1 Intended use. The canister assembly is intended for use as a component of the M259, warhead and XM264 warhead.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Preproduction:

(1) Time allowed for suppliers submission of samples for Government test and evaluation after award of contract.
(2) Destination of samples when testing is performed by the Government.
(3) Packing and shipping instructions when applicable.
(4) Time required for the Government to notify the suppliers whether to proceed with production.

6.3 Caution. Due to the hazards inherent in testing with high pressures, suppliers are advised to take all necessary precautions to protect personnel while testing.

6.4 Supersession data. This specification supersedes Chemical Systems Laboratory Purchase Description EA-C-1158, 14 September 1977.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1340-A545

☆U.S. GOVERNMENT PRINTING OFFICE: 1978-703-122/3562

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-C-51448 30 JAN 78

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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